From Blind Estimation to Frequency-Domain Control of a Brain–Machine Hand Interface

GG4 — Neural Data Analysis with Adaptive State Estimation | yhl63

Abstract

This report presents the evolution of our approach to modelling and controlling an unknown brain–machine interface (BMI), progressing from a blind linear–Gaussian estimation framework to a frequency-domain control paradigm informed by system structure and empirical observations. Analysis of the BMI decoder and empirical frequency response revealed that the brain's best linear approximation is well-described by a first-order low-pass response, even though a substantial nonlinear component remains (input–output coherence ∼0.6); this explains the poor high-frequency model fidelity and motivates a shift away from time-domain control. The controller exploits only the coarse property that a tone at one frequency yields a latent oscillation at the same frequency, and closes a position loop to correct the rest, making it robust to the nonlinearity it cannot capture. Early LQG/LQI approaches, based on next-step prediction and standard LQR objectives, proved ineffective as they do not account for the system's reliance on oscillatory inputs. Exploiting linearity and stability, we reformulate control in the frequency domain, enabling open-loop frequency matching and eliminating the need to track fast neural dynamics. By leveraging time-scale separation, we close a single outer loop on position using steady-state responses. The resulting controller achieves consistent performance within 6 cm, with limitations directly attributable to low-pass attenuation at higher frequencies.

1 Problem Definition and Introduction

The system is composed of three principal modules. The neural dynamics from a "brain" are estimated as a linear Gaussian state-space model (the assumptions of this model will be discussed later), which maps a two-dimensional input signal $u \in [0,1]^2$ to a 16-dimensional neural activity vector $y \in \mathbb{R}^{16}$. A brain–machine interface (BMI) then projects y onto a two-dimensional latent state (x_1, x_2) via a linear transformation. This latent representation serves as input to a recurrent neural network (RNN) that produces muscle activation signals for a biomechanical arm model comprising planar shoulder and elbow joints actuated by four antagonistic muscles.

$$u \rightarrow \underbrace{\text{brain}}_{\text{LGSSM}} \rightarrow y \in \mathbb{R}^{16} \rightarrow \underbrace{\text{decoder}}_{(x_1,x_2)} \rightarrow \underbrace{\text{muscle head}}_{\text{nonlinear}} \rightarrow 4 \text{ muscles} \rightarrow \underbrace{\text{2-link arm}}_{\text{integrator}} \rightarrow \text{hand},$$

Our final objective is to steer a simulated hand to a Cartesian target using a closed-loop controller. Our approach is based on a sequence of insights and design choices, motivated by the structure of the BMI and the empirical data obtained through the Brain model. We explore a series of exploration approaches following our interim report and the insights they provided, leading to our final controller design and its evaluation.

2 From blind estimation to active probing

Week-2 starting point. The interim work solved a more difficult estimation problem by attempting to recover both the latent state and an unknown input from observations alone, with no access to u. We modelled the brain as a linear-Gaussian state-space model and closed the problem with an augmented-state construction: the input is stacked onto the latent, $z_k = [x_k^\top \; u_k^\top]^\top$, under an AR(1) random-walk prior $u_{k+1} = u_k + \eta_k$, giving the block-triangular dynamics $z_{k+1} = \left[\begin{smallmatrix} A & B \\ 0 & I \end{smallmatrix}\right] z_k + \tilde{w}_k$, $y_k = [C \; D] z_k + v_k$, such that a single Kalman/RTS smoothing pass returns the state in its leading n and the input in its trailing m coordinates (where n and m are the dimensions of the latent and input spaces, respectively), while EM (closed-form M-step) fits the matrices. We warm-started the EM from PCA of the observations (C from the leading n singular vectors, A from one-step least-squares on the PCA scores), with a few random restarts against local optima. We discussed the limitations of folding the input into the latent : the AR(1) increment prior biases the recovered input toward smoothly-varying signals and represents sharp or impulsive inputs only approximately, and the state/input split is identified only relative to that prior (and the exogeneity constraint of the zero lower-left block)

Known inputs with a black-box model. In Weeks 3–4 we command u, so identification becomes well-posed and enables probe design. Under amplitude constraints ($[0, 1]$ clipping), the most informative inputs lie at the bounds: we employ multi-level, piecewise-constant signals with holds across timescales $1, 4, 16, 64$ as long holds concentrate energy at low frequencies to excite slow dynamics within finite data, while short holds probe faster modes; multiple amplitude levels also expose nonlinearities. This approach outperforms impulses (energy-limited under clipping) and Gaussian noise (high crest factor), and aligns with D-optimal design by maximising $\det M$. For the observed low-pass system, concentrating power in the low-frequency band where control authority is needed yields the most informative data.

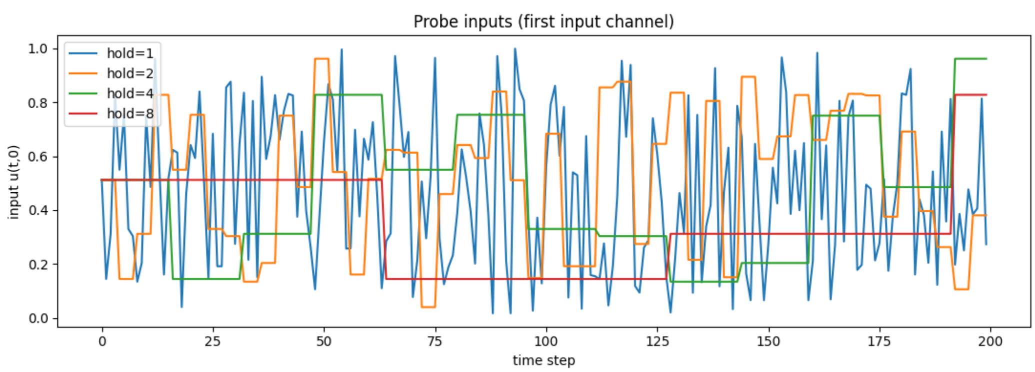

Figure 1: The persistently-exciting probe: multi-level, piecewise-constant signals with holds across timescales.

Knowing the inputs removes the need for the input prior and the augmentation, allowing us to improve on our previous blind estimation by seeding EM with an input-aware subspace estimate via the block-Hankel matrix. The leading left singular vectors of a block-Hankel matrix of the output data, after projecting out the future-input dependence (a MOESP-style step), give a consistent estimate of the extended observability matrix, from which C and A follow by shift-invariance. With (A, C) fixed, (B, x_0) follow from a linear least-squares regression of the observed outputs onto the simulated responses to unit impulses in each input channel and unit initial-state directions. This data-driven warm start, enabled by the persistently-exciting probe, seeds B (which PCA cannot) and remains usable when the number of latent variables exceeds the number of outputs n¿p (which PCA also cannot), placing EM in a markedly better basin. An ambiguity remains, however, in the choice of Hankel window-length, which we determine based on the number of latent dimensions we choose to fit and the number of held-out predictive score.

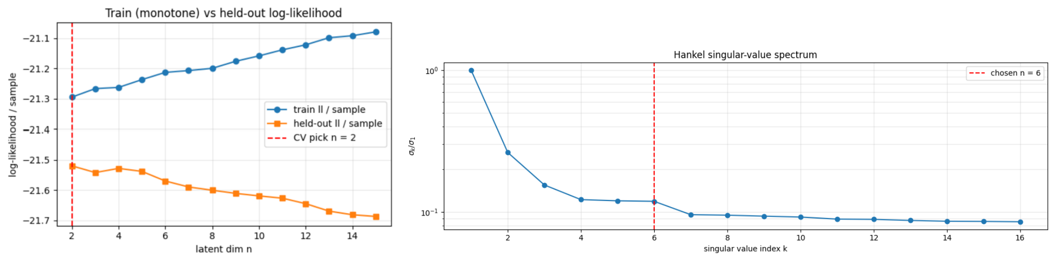

Figure 2: Model order selection diagnostics. Held-out predictive score (left) and singular-value spectrum of the block-Hankel matrix (right) both suggest retaining $n = 6$ modes, with a clear gap after $n = 2$ and a noise floor from $n \approx 6$ onward.

Held-out predictive scoring, repeated over independent simulated trials rather than data-recycling cross-validation, places the model order on a plateau at n=2, consistent with m=2 inputs ($\leq$ 2 controllable directions) and a 2-D decoder readout. From observing the trained-log-liklihood (in blue) we see that the model performs well on the training data but begins to overfit as the number of modes increases, while the held-out log-likelihood (in orange) decays after n=2. The same block-Hankel matrix offers a complementary, structural check via its singular-value spectrum: σ_k/σ_1 drops sharply to k=2, declines gradually through k≈3–6, and flattens into a noise floor from k≈6 onward, the soft-knee signature expected under measurement noise, where the floor marks directions blending into noise. The two diagnostics ask different questions: the spectrum asks whether a mode is distinguishable from noise in this probe at all, while the held-out score asks whether estimating that mode's parameters from this much data improves free-run prediction. We read the gap between n=2 and the n≈6 floor as weakly-excited modes that are real but not reliably identifiable from a probe of this length, and retain n=6 as the operating order — trading predictive parsimony for the controllability margin the downstream LQR/LQI design relies on.

3 Initial control: setpoint LQG/LQI, and why it changes

Prior to recieving the BMI, our first controllers were based on setpoint regulation using standard model-based methods. Setpoint regulation picks a feasible target $u^\star = G^+ y^\star$, $x^\star = (I - A)^{-1} B u^\star$ (G is 16×2, so $y^\star$ is met only in least squares) and applies the steady-state LQG law

$$u_t = u^\star - K(\hat{x}_t - x^\star),$$

with K the infinite-horizon LQR gain (control DARE) and $\hat{x}$ from a steady-state Kalman filter (estimator DARE) that re-anchors the state each step from the innovation $y - C\hat{x}$. The feedforward $u^\star$ is responsible for holding the setpoint, without it the law reduces to $-K\hat{x}$, which vanishes at zero error. However, it is still not offset-free as $u^\star$ uses our model's DC gain, so model mismatch would still leave a residual error. **LQI** removes it with a pure integral servo $u_t = -K_x \hat{x}_t - K_i \xi_t$ that integrates the measured tracked-output error, $\xi_{t+1} = \xi_t + (H y_t - z^\star)$, driving it to zero at DC (with anti-windup under the $[0, 1]$ clip) at the cost of a slower approach.

Under-actuation ($m = 2$ inputs, $p = 16$ outputs) restricts the steady-state reachable set to $\operatorname{im} G$ (rank ≤ 2), making an arbitrary 16-D output target is unattainable and can only be resolved via its least-squares projection $u^\star = G^+ y^\star$. Tracking therefore requires compressing the control target into a lower-dimensional variable $z = Wy$ ($\dim \leq m$). Unknown to us at the time of implementing this controller, this dimensional reduction is inherently handled by the downstream architecture, the 16 plant outputs feed directly into an affine decoder designed to isolate the two latent rhythms, making $z = (x_1, x_2)$ the definitive tracked variable. By routing the high-dimensional plant through this downstream decoder, the end-to-end latent transfer function collapses into a square 2×2 map:

$$H_z(e^{j\omega}) = WC(e^{j\omega}I - A)^{-1}B$$

B ecause this map is square, it is trivial to invert—effectively bypassing the non-square under-actuation of the physical plant and providing the core operator utilized by the frequency-based framework in the next section.

4 System identification and why it struggles

We empirically measured the brain's frequency response using a periodic multisine (averaging periods to estimate free noise) and compared it to the identified model (Fig. 3). Three structural findings shape our control design:

1. Model fidelity is restricted to the dominant direction at low frequencies. With m=2 inputs, the response has two principal gains, σ_1 and σ_2. The dominant gain σ_1 matches well at low frequencies. However, σ_2 is under-represented by the model almost everywhere, and the relative reconstruction error climbs toward 1 at high frequencies where the brain transmits little signal.

2. A coherence ceiling reveals unmodeled nonlinearity. Input–output coherence never approaches 1, even at peak SNR. This fixed ceiling is the classic fingerprint of a nonlinear stack: a specific fraction of the output is not linearly explainable, and no increase in latent dimension removes it. While period-averaging lifts the SNR-limited deficit at high frequencies, the low-frequency coherence ceiling remains structural and intractable.

3. The surrogate is strictly a best-linear-approximation. Because complex exponentials are the eigenfunctions of an LTI map, the transfer function $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ is invariant to internal coordinate transformations ($x \to Tx$). Internal coordinates are never uniquely identifiable, but H is. Therefore, the EM-fitted model and the model-free empirical transfer function should coincide; where they do not (Fig. 3), the gap is model error rather than an artefact of coordinates. Against a nonlinear plant, this gauge-invariant matching simply extracts the best linear approximation at the probed amplitude.

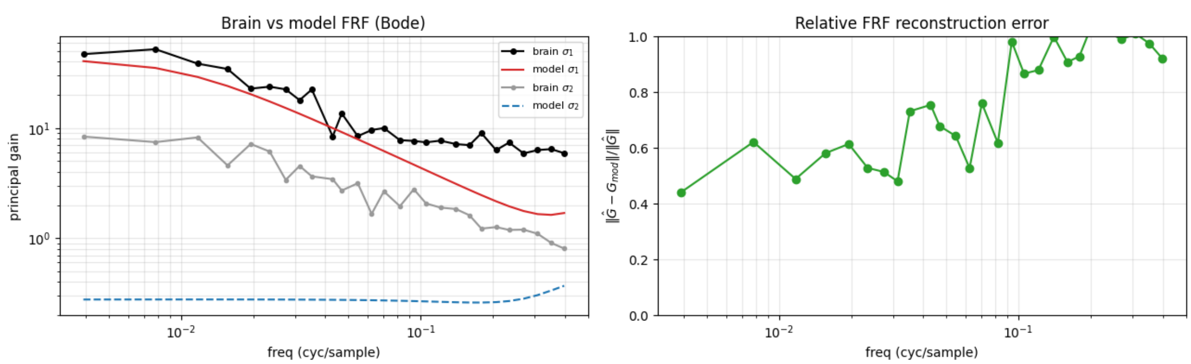

Figure 3: Empirical (black/grey) vs. identified-model (red/blue) principal gains (left) and relative reconstruction error (right). σ_1 is captured at low frequency; σ_2 is badly under-modelled and the error grows with frequency, where coherence collapses — the regime the controller must avoid.

5 The brain is a first-order low-pass system

Across all seeds, the identified state matrix A is dominated by a single pole pair at $|\lambda| \approx 0.93$–0.95. Fitting the principal gain $\sigma_1(f)$ to a first-order low-pass filter $|k/(1 - ae^{-j\omega})|$ yields an exceptional fit ($a = 0.940$, $R^2 = 0.998$; Fig. 4), revealing a -3 dB cutoff near $f_c \approx 0.012$ cyc/sample followed by a steep

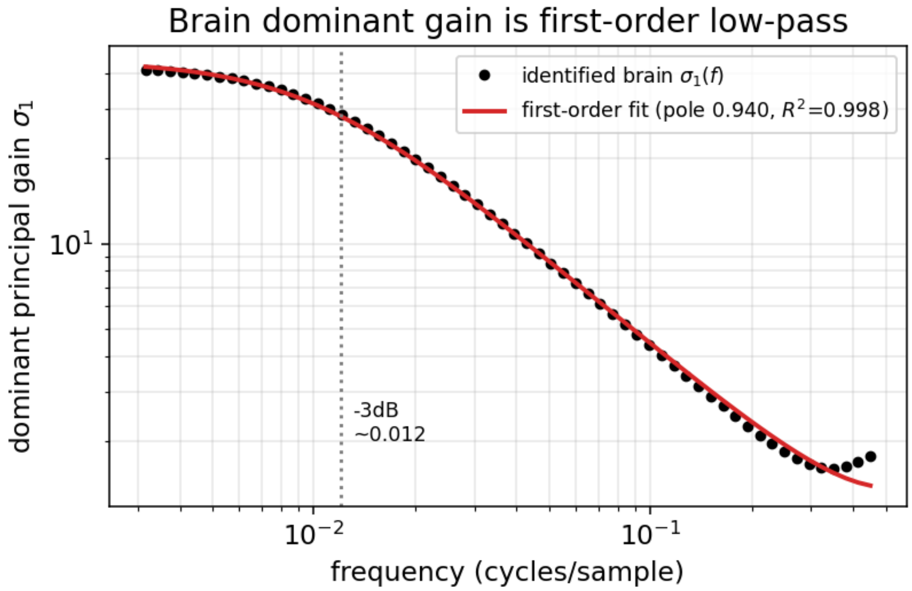

Figure 4: The identified brain's dominant gain $\sigma_1(f)$ (markers) against a first-order low-pass fit (red, R^2=0.998, pole 0.94). The muscle frequencies sit far above the -3dB cutoff, on the roll-off.

23× roll-off. This physical limitation dictates our downstream performance, which will be referenced in the controller's dominant failure mode in §8.

6 The BMI structure forces a frequency-domain view

Inspecting the interface reframed our approach. The decoder taking the 16-D neural activity y to the 2-D latent state z is affine, so it drops into any linear controller unchanged. The muscle head, however, is nonlinear, stateful, and rhythm-coded: muscle selection is the band-energy of x_1's recent history through four fixed band-pass filters, in other words, the oscillation frequency of x_1 chooses the muscle while power gates on x_2 clearing a threshold with persistent evidence. Activation is power × selector, and the arm integrates muscle differentials into joint angular velocities, with a dry-friction deadzone. As a result, we find that a constant x_1 produces no band energy and hence zero selection, to move the hand, x_1 must oscillate, leading to the interpretation that the natural language for the inner control problem is the brain's frequency response. This is what turned the investigation from time-domain regulation toward frequency matching.

7 The final controller: frequency-multiplexed joint servo

Our approximation of the brain being a stable, linear, low-pass system lets us replace the inner regulation problem with open-loop frequency synthesis, letting the only feedback live at the outer, position level, significantly simplifying our design.

Small transients allow us to bypass continuous neural observation. The brain's dominant pole (0.94) yields a transient time constant of $\tau \approx 17$ steps and a settling time of ∼50 steps, far shorter than the $T = 1000$ control horizon and the tens-of-steps cadence of hand corrections. Following this brief initial transient, the latent oscillations reach a steady state fully characterized by the frequency response. This allows the controller to inject the required spectral content open-loop, ignoring the fast neural transients entirely.

The frequency synthesis loop was built in three stages :

1. Naive two-tone baseline: Initial exploration assumed symmetric dynamics, driving x_1 with a high-frequency select tone to activate the downstream band-pass filters, and x_2 with a slow power tone to modulate muscle activation magnitude. This architecture failed due to inner-channel cross-talk: the select tone leaked into both latents, causing x_2 to transiently dip below its critical activation threshold. Because the muscle head recovers power slowly after a threshold violation, these dips collapsed continuous actuation into a single, un-retriggerable "fire-once" transient.

2. Input-direction nulling: To decouple the channels, we inverted the 2×2 transfer function matrix H_z to actively null the select frequency leakage along the x_2 coordinate. While mathematically well-conditioned, this AC-power strategy required centering the x_2 oscillation. Shifting the state to center this trajectory consumed the strict $[0, 1]$ input headroom, causing severe control signal saturation. This clipping reintroduced the exact harmonic distortion and cross-talk the nulling matrix was designed to eliminate.

3. Inherent DC exploitation (Final): Direct inspection of the decoder network's weights and biases resolved this bottleneck by revealing a fundamental structural asymmetry: x_2 operates purely as a magnitude-dependent power throttle, and its natural resting DC offset ($\approx +1.8$) already sits securely above the muscle activation threshold. Consequently, oscillating x_2 is not only unnecessary but counterproductive. By abandoning the active power tone entirely, we compress the control problem to a single dimension: emitting a small select tone on x_1. This tone is large enough to breach the selection threshold, yet small enough that its residual leakage never drives x_2 below its resting threshold, avoiding input saturation entirely and turning a structural constraint into a control asset.

Outer servo. Muscles 0/1 drive the shoulder and 2/3 the elbow, and each joint's velocity depends only on its own pair, so control decouples into two independent inverse-kinematics joint servos. Using the known arm geometry (link lengths etc.), we uniquely resolve the multi-pose ambiguity in current-pose-estimation by strictly tracking the continuous, minimum-displacement branch relative to the current joint angles with an L2 norm on the joint error. As for the target-pose, we choose the pose where the elbow angle is positive, which recruits the strong, low-frequency elbow+ muscle rather than the weak, high-frequency elbow− muscle, motivated by our discussion on the low-pass structure of the brain (more in §8). With the target pose fixed, each joint error emits a select tone of amplitude $a = \text{clip}(k|e|, a_{\min}, a_{\max})$, and inside a deadzone the tones drop and stiction holds the pose. To counteract the remaining asymmetry in the brain's high-frequency attenuation, the elbow's command is scaled by an empirical scalar multiplier ($a \leftarrow a \cdot \alpha_{\text{elbow}}$) to equalize joint authorities. Inside a tracking deadzone, the tones drop to zero and physical stiction holds the pose. The four select frequencies and the joint map are calibrated once offline (Fig. 5).

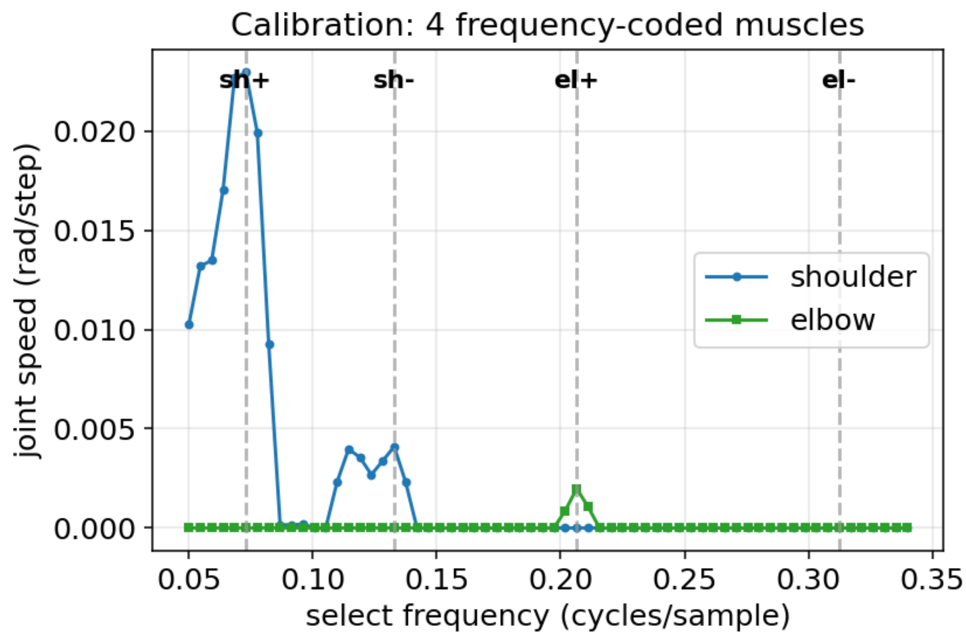

Figure 5: Offline calibration: sweeping the select frequency and measuring steady joint velocity exposes four band-pass muscle peaks. Higher-frequency peaks are weaker — the low-pass roll-off of §5 seen at the actuator level.

8 Evaluation methodology and results

Methodology. All controllers are run through the group-standard harness for commensurability: an identical polar target grid (radii $\{20, 38, 55\}$ cm $\times$ 10 angles), a fixed brain seed, horizon T=1000, and six metrics chosen to expose distinct behaviours : final distance, time-to-threshold, steady-state distance (settling), the shoulder/elbow decomposition, control effort, and a spatial heatmap for visualisation.

On the grid (Fig. 6) the final distance is 32.3 ± 33.4 cm, shoulder error $19.3 \pm 23.1°$, elbow error $53.1 \pm 63.0°$, time-to-threshold 401 ± 211 steps, effort near-constant (617 ± 23). Every standard deviation rivals its mean: the average hides a sharply *bimodal* outcome.

A directional failure, predicted by the low-pass model. We notice that the upper half of the reachable space is reached to a few cm; while the lower half misses by $\sim$60 cm, dominated by elbow error. The two halves recruit different members of each antagonist pair, and measured joint authority falls monotonically with frequency: shoulder+ (0.073) gives 9.7 rad of travel, shoulder− (0.133) 2.0, elbow+ (0.207) 2.0, elbow− (0.312) only 0.9, following the ordering of the low-pass roll-off we noticed from (§5). The highest-frequency muscle, elbow−, is near-dead, so the lower half is reach-limited. The flat effort across radius (Fig. 6f) confirms an actuation limit rather than under-driving.

9 Strengths, limitations, alternatives, and future work

Strengths : The controller exploits rather than fights the decoder (frequency selection, free power), needs no online model inversion, generalises across brains, and its failure is interpretable and traceable to one measured cause.

Limitations : The directional failure stemming from the brain's low-pass roll-off as discussed severely under-actuates the high-frequency elbow muscle. This attenuation is compounded during simultaneous multi-joint movements, as the shared [0, 1] input budget forces the already starved elbow

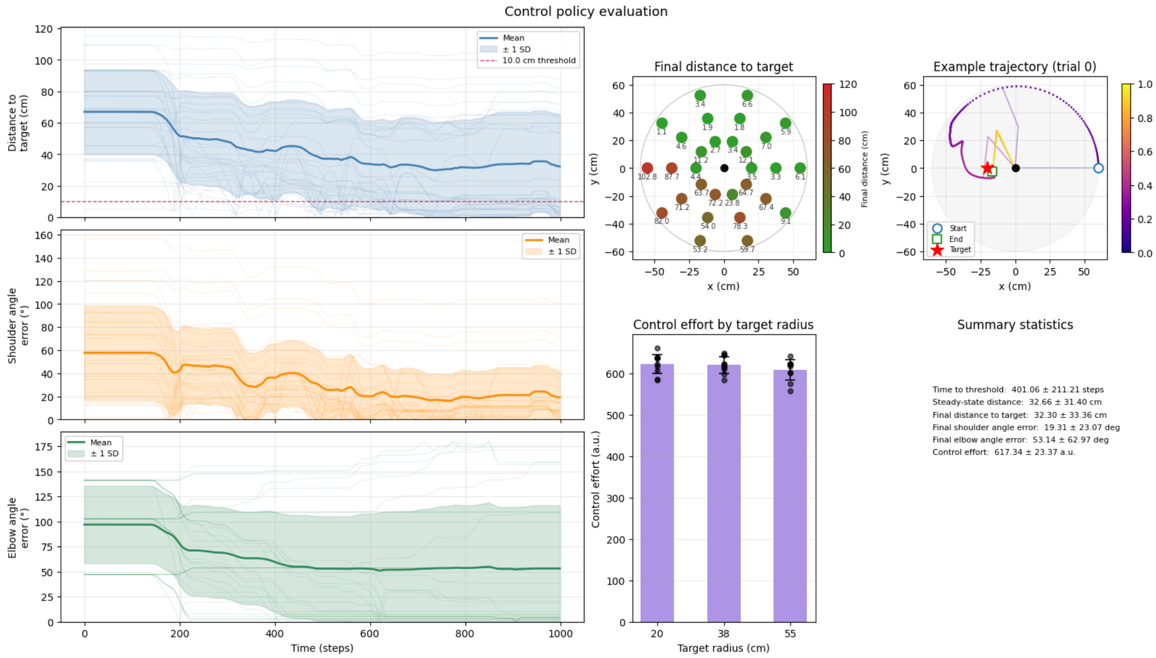

Figure 6: Standardised evaluation (seed 0, T=1000). (a) Distance: a ∼150-step power ramp then a plateau with a wide band (bimodal). (b,c) The shoulder error falls but the *elbow* error stays high. (d) The final-distance heatmap is sharply *directional*: a reachable upper/right region ($\leq$ 12 cm) and a red lower/far-left shortfall (53–103 cm). (e) Example reach. (f) Effort is flat across radius — the failing targets spend effort without converging.

to compete with the shoulder for limited control authority. Furthermore, the asymmetric dynamics of the muscle head characterized by a slow power ramp and a fast exit establish a strict resolution floor. Because the smallest effective stimulation burst often exceeds the micro-correction required near the target, the controller tends to overshoot and "hunt" around the setpoint. As a result, ultimate steady-state precision is inherently bounded by the arm's friction deadzone, since the controller cannot easily punch through stiction without triggering this overshoot limit-cycle.

An attempted improvement: kinematic redundancy As discussed in our controller design, almost every target is reachable in *two* arm configurations, elbow-up ($+|\theta_{\mathrm{el}}|$) and elbow-down, and the controller's default chooses the branch nearest the home pose based on the L2 error, which for the lower workspace demands the weak, high-frequency elbow− muscle. Forcing the elbow-up branch reaches the same Cartesian point with the strong, low-frequency elbow + muscle, which is less affected by the brain's low-pass roll-off.

target IK branch	overall	lower half	within 15 cm
default (nearest home)	34.7 cm	61.6 cm	46%
elbow-up (strong muscle)	**10.3** cm	**12.7** cm	**79**%

Tracking the differences between the two applications, the lower half collapses from 61.6 to 12.7 cm and reliability nearly doubles, confirming the diagnosis. Routing the task onto muscles in the brain's well-actuated band by applying our frequency-domain insight applied at the kinematic level provided signifcantly improved results. **Future work :** Beyond the attempted improvement to resolve the attenuation of high-frequency muscles, the limitation of the resolution floor which contributed to the hunting behaviour near the target could be mitigated by implementing a hybrid control strategy that switches to a more conservative, low-frequency stimulation pattern as the hand approaches the target, reducing overshoot and improving steady-state precision. Additionally, exploring adaptive control techniques that can learn to compensate for the brain's low-pass characteristics over time may further enhance performance, especially in the presence of variability across different brain seeds or changes in the system dynamics.

10 Conclusion

The arc is one of progressive simplification driven by understanding the plant, built upon explorations of system identification both blind and informed; the rhythm-coded decoder forced a frequency-domain view; and identifying the brain as a first-order low-pass system collapsed the inner control problem to open-loop frequency matching with a single outer position loop, justified because the controller horizon dwarfs the brain's transient, as well as predicting the directional failure of the under-actuated elbow joint. Our final controller achieves reasonable performance for its simple model-free design, with limitations directly attributable to the brain's low-pass structure. Future work will explore more sophisticated control architectures that can better navigate the trade-offs imposed by the brain's dynamics and the muscle head's nonlinearities, as well as investigating adaptive strategies that can learn to compensate for these limitations over time.